ตัวอย่าง · ตัดข้อมูลลูกค้าและราคาออกแล้ว · Sample: customer details and prices removed

Solar ESS Co. Ltd. · บริษัท โซลาร์ เอสส์ จำกัด
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PROPOSAL · ON-GRID SOLAR ROOFTOP

15 kWp On-Grid Solar Rooftop System

Prepared for **a subdistrict municipality** , office building · 12 October 2026

TURNKEY PRICE

— THB before VAT

— THB including VAT · built to the 15 kWp TOR

— THB per watt
15,600 W installed

15.6 kWp
24 × LONGi Hi-MO X10 650 W

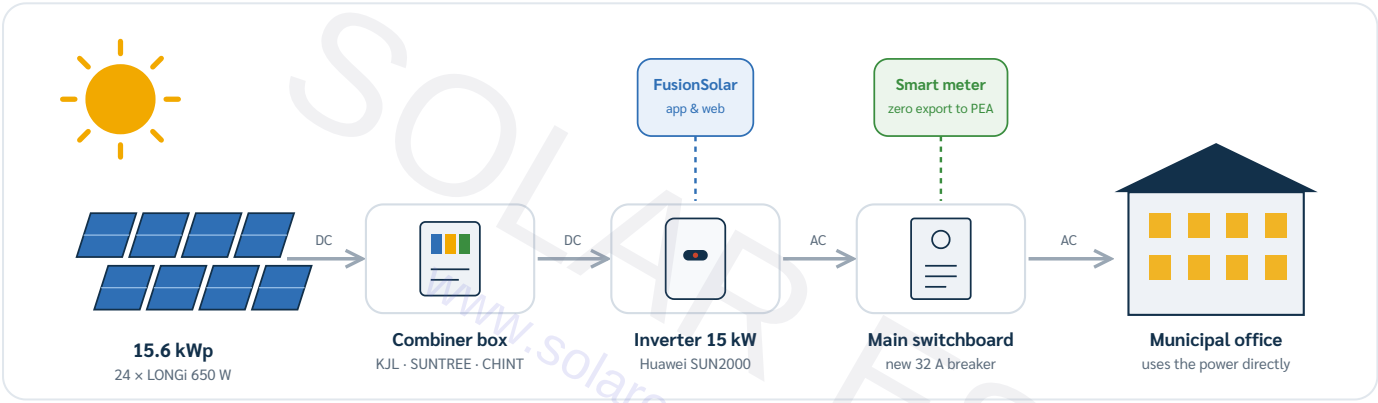
15 kW
Huawei SUN2000-15KTL-M5, 3-phase

20,850 kWh
production in year 1

4.7 years
payback on — THB

1.35 M THB
net saving over 25 years

≈ 9 t CO_2
avoided every year



Solar ESS will survey, design, supply, install and connect the system, handle the PEA application, and maintain it for two years. The inverter feeds the office directly; a smart meter keeps export to the PEA grid at zero.

1 About Solar ESS

Solar ESS Co. Ltd. (บริษัท โซลาร์ เอสส์ จำกัด) · "Solar Energy for Sustainable Solution" · Tax ID 0575566002051

We design and install turnkey solar systems from our head office in Chiang Rai, with branches serving the south of Thailand.

Offices
Head office Chiang Rai · branches in Krabi, Koh Samui (Surat Thani) and Khanom (Nakhon Si Thammarat)

Services
Survey, design, supply, installation, PEA application with a signed SLD, cleaning and maintenance

Systems
On-grid, hybrid and off-grid, from homes to commercial and institutional buildings

Equipment
Huawei inverters on most of our sites; LONGi panels; SUNTREE, CHINT and ABB protection; KJL enclosures; Yazaki and LINK cable

Solar ESS Working Group	Watcharote Iamklam	Anucha Intongead	Artitaya Khuanphet
Chief Executive Officer	Chief Technical Officer · licensed electrical engineer (ภฟท. 44981)	Chief Operation Officer	Chief Legal Officer

Reference sites can be viewed on our website, www.solaress.org.

2 Proposed system

A 15.6 kWp array on one Huawei 15 kW inverter, connected to the existing switchboard with zero export.

PV array
24 × LONGi Hi-MO X10 650 W (LR7-72HVH-650M), 2 strings of 12 = 15.6 kWp

Inverter
Huawei SUN2000-15KTL-M5, 15 kW, 3-phase 400 V, 2 MPPT

Grid connection
To the existing MDB through a new 32 A CHINT breaker; zero export

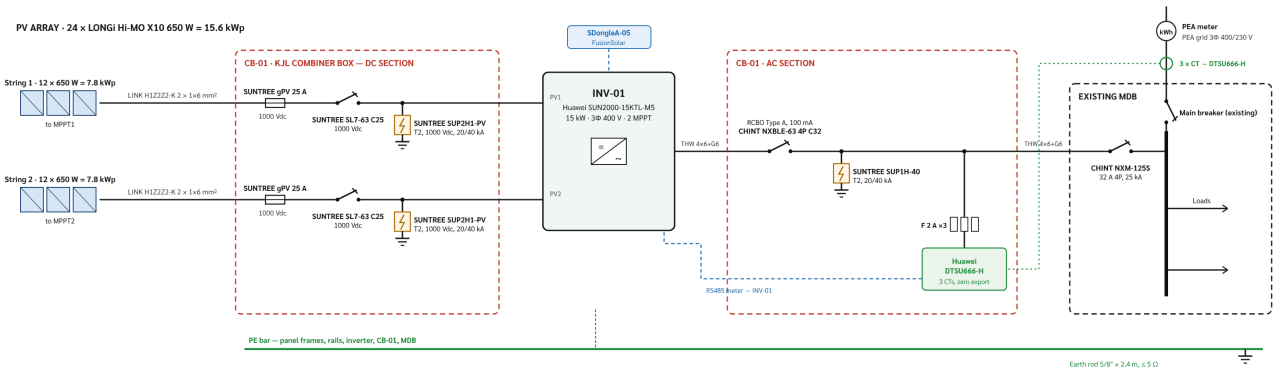
Monitoring
Huawei SDongleA-05 to the FusionSolar app and web portal

Combiner box
KJL steel IP65 with SUNTREE DC and CHINT AC protection

Mounting and earthing
Aluminium rails, stainless fittings, frames bonded to an earth rod

Single line diagram

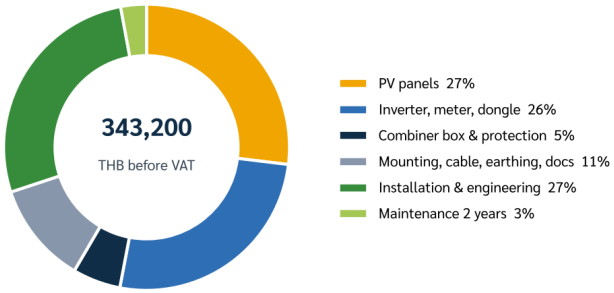
Each string runs through its own SUNTREE fuse, DC breaker and surge protector to one MPPT; the inverter output passes a CHINT RCBO and AC surge protector to the MDB. The DTSU666-H meter reads the incoming supply through 3 CTs to hold export at zero.



3 Bill of materials and price

– THB before VAT, – THB per watt: – THB materials and – THB labour. VAT 7% applies to materials only.

#	Item	Brand / model	Qty	Unit (THB)	Amount (THB)
PV array					
1	PV module 650 W, HPBC 2.0, delivered to site	LONGi Hi-MO X10 LR7-72HVH-650M	24	–	–
Inverter and monitoring					
2	On-grid inverter 15 kW, 3-phase, 2 MPPT	Huawei SUN2000-15KTL-M5	1	–	–
3	Monitoring dongle (WLAN-FE)	Huawei SDongleA-05	1	–	–
4	Smart meter 3-phase with 3 CTs, zero export	Huawei DTSU666-H	1	–	–
Combiner box and protection					
5	Combiner box, steel IP65, lockable	KJL	1	–	–
6	DC fuse holder 1100 Vdc + gPV fuse 25 A	SUNTREE SRD-30 + SRF-30	4	–	–
7	DC MCB C25, 1000 Vdc	SUNTREE SL7-63 4P	2	–	–
8	DC SPD Type II, 1000 Vdc, 20/40 kA	SUNTREE SUP2H1-PV 3P	2	–	–
9	RCBO 4P C32, Type A, 100 mA	CHINT NXBLE-63	1	–	–
10	AC SPD Type II 4P, 385 Vac + backup MCB	SUNTREE SUP1H-40 + SCB8-63	1	–	–
11	Meter fuse holder 1P + 2 A fuse	SUNTREE SRD-30	3	–	–
12	Terminals, wiring duct, DIN rail, labels	KJL	1 lot	–	–
13	MCCB 4P 32 A, 25 kA, PV feeder in the MDB	CHINT NXM-125S/4P	1	–	–
Mounting, cable, earthing and documents					
14	Aluminium rails, L-feet, clamps, stainless bolts	Aluminium + SUS304	24 panels	–	–
15	PV cable H1Z2Z2-K 1×6 mm², 1.5 kV DC	LINK	200 m	–	–
16	MC4 connector pair, 1500 Vdc	SUNTREE	6	–	–
17	AC cable THW 1×6 mm² (L1, L2, L3, N, PE)	Thai Yazaki	150 m	–	–
18	Earth rod 5/8" × 2.4 m, earth cable and clamps	KUMWELL + Thai Yazaki	1 lot	–	–
19	Ground test box	KUMWELL GYPTB	1	–	–
20	Metal trunking with cover, conduit, fittings	KJL	1 lot	–	–
21	As-built drawings (A3 × 2, DWG + PDF on 3 flash drives), Thai manual × 2	Solar ESS	1 lot	–	–
Labour (not subject to VAT)					
22	Installation, testing and commissioning, 15,600 W at – THB/W	Solar ESS	15,600 W	–	–
23	Engineering, shop drawings, PEA application and SLD signed by a licensed engineer	Solar ESS	1 lot	–	–
24	Maintenance for 2 years, panel cleaning once a year	Solar ESS	2 years	–	–



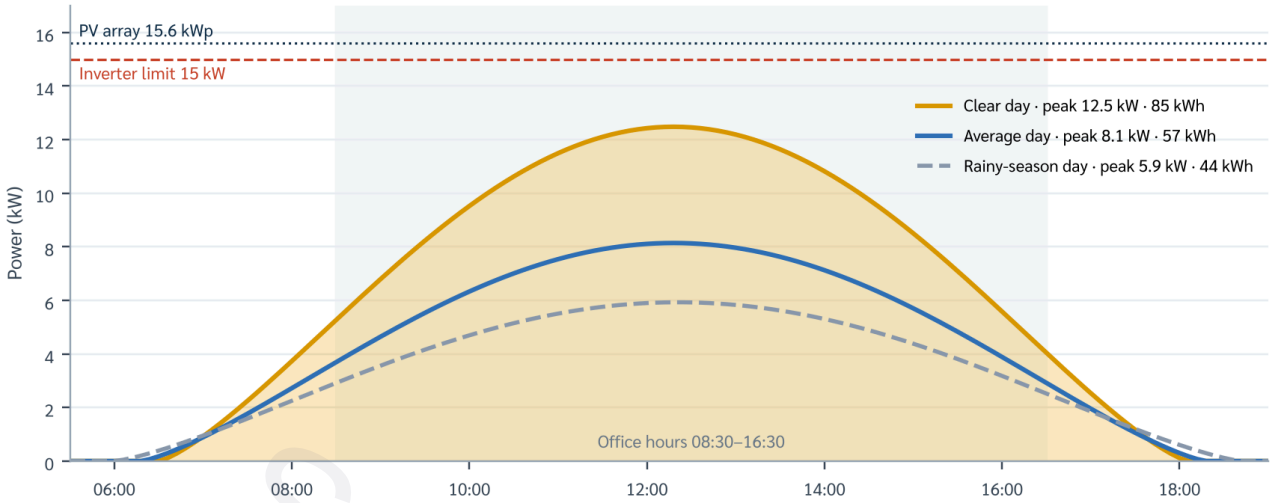
Materials	–
Labour (no VAT)	–
Total before VAT · – THB/W	–
VAT 7% on materials	–
Grand total including VAT	–

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4 Energy production

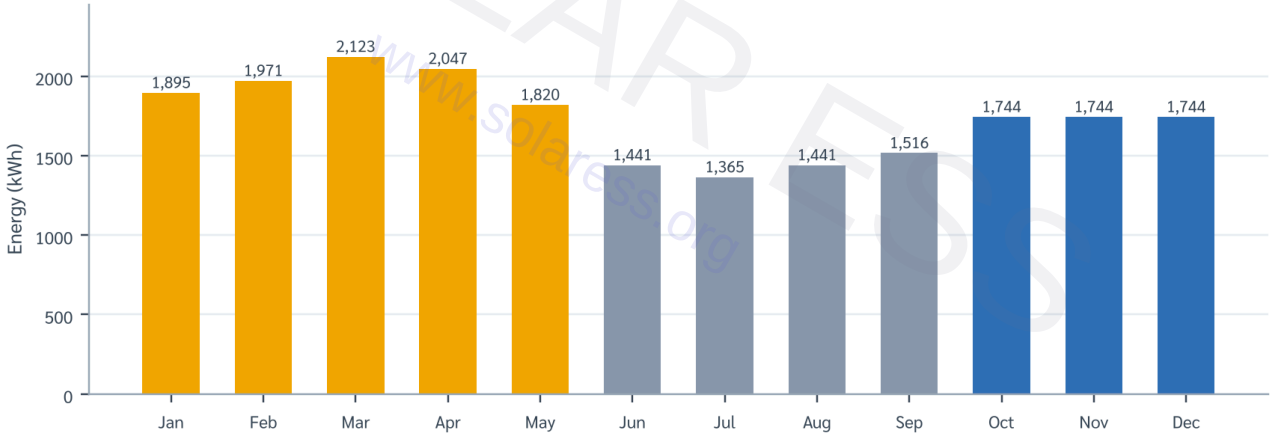
About 20,850 kWh in year 1. Even on a clear day the array peaks at 12.5 kW, below the inverter's 15 kW limit, so nothing is clipped.

Through the day, against panel and inverter capacity



Most output falls inside office hours, when the building uses it directly. DC-to-AC ratio 15.6 kWp : 15 kW = 1.04.

By month, year 1



March and April produce most (about 2,100 kWh each); July produces least (about 1,370 kWh) because of monsoon cloud. Typical Chiang Rai profile scaled to 1,350 kWh per kWp, less 1% first-year degradation.

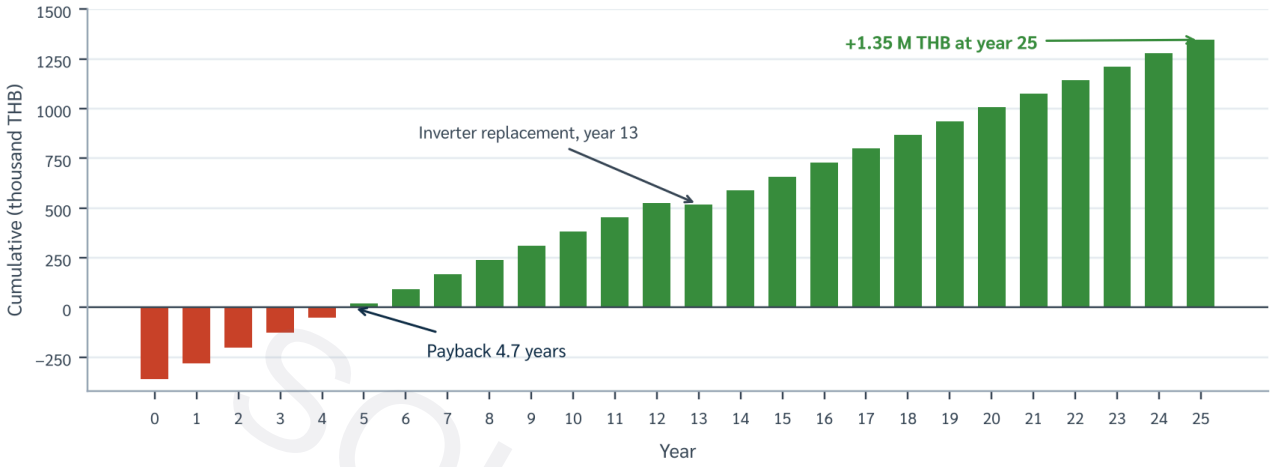
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5 Return on investment

The – THB investment pays back in about 4.7 years and returns 1.35 million THB over 25 years.

4.7 yrs	20.2%	79,749 THB	≈ 9 t
payback	IRR over 25 years	saving in year 1	CO_2 avoided a year

Cumulative cash flow, base case



Assumptions

Investment	– THB incl. VAT (not reclaimable)
Production	1,350 kWh/kWp; –1% year 1, then –0.4%/yr
Used in office	85% (rest held back by zero export)
Electricity price	4.50 THB/kWh incl. Ft and VAT, flat
Running costs	5,000 THB/yr from year 3; inverter 79,500 THB in year 13

Sensitivity

Case	Payback	25-yr saving	IRR
70% used	5.8 yrs	1,010,944	15.9%
85% used (base)	**4.7 yrs**	**1,346,396**	**20.2%**
100% used	4.0 yrs	1,681,849	24.3%
85%, tariff +2%/yr	4.5 yrs	1,870,950	22.3%

The share of output the office uses is the figure to confirm: the office closes at weekends, so a 15-minute load profile from the MDB would firm up the base case. CO_2 at the Thai grid factor of about 0.5 kg/kWh.

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6 Scope of work and schedule

Delivered turnkey within 90 days of contract signing, as the TOR requires; site work takes about 20 days.

1. **Survey and design**
   Roof, structure load and MDB survey; calculations and shop drawings signed by a licensed engineer within 7 working days
2. **Supply**
   All equipment new and unused, with manufacturer certificates
3. **Installation**
   Rails, panels, combiner box, inverter, trunking and earthing; roof fixings sealed against leaks
4. **Connection**
   New 32 A breaker in the existing MDB; any shutdown agreed 7 working days ahead, 09:00–16:00
5. **Testing**
   Commissioning to IEC 62446-1, zero-export setting and FusionSolar monitoring
6. **PEA approval**
   Grid-connection application handled by Solar ESS until approved for use
7. **Handover**
   As-built drawings, Thai O&M manual and on-site training for municipal staff

7 Warranty and maintenance

All warranties meet or exceed clause 11 of the TOR.

- **PV modules, product**
  At least 10 years, per the LONGi warranty certificate
- **PV modules, power output**
  At least 25 years linear, not less than 80% output
- **Inverter**
  10 years, claims handled directly by Huawei Thailand
- **Workmanship and materials**
  2 years from handover
- **Repair response**
  On site within 7 days of notice; repaired part re-warranted for 2 years
- **Maintenance**
  Cleaning and electrical safety check once a year for 2 years, included

8 Terms

- **Payment:** one payment after completion, PEA approval for use and acceptance by the inspection committee (TOR clause 9).
- **Price:** Thai baht; VAT 7% on materials, labour not subject to VAT; valid until 11 November 2026.
- **Included:** PEA connection application, signed SLD, shop drawings, as-built drawings, manual and training.
- **Not included:** roof repairs beyond the panel fixings, MDB replacement, and an application to sell power to PEA (2,140 THB if requested).
- **Delivery:** within 90 days of contract signing.

Proposed by
Solar ESS Working Group
Chief Executive Officer, Solar ESS Co. Ltd.
Date _____

Accepted by
a subdistrict municipality

Date _____

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